

MOTHER EXO and graphene in robotics

Building a humanoid world model on open foundations — the stages, the research, and the engineering.



— Abstract

Humanoid robotics is not gated by actuators. It is gated by experience — and experience is expensive, slow and dangerous to collect on real hardware. This paper describes how MSAI Robotics is closing that gap: an open-source mechanical baseline, a set of modules designed in-house where the baseline runs out, a graphene material programme with its confidence levels stated rather than implied, and a physics digital twin that lets a world model accumulate experience faster than a workshop can.

It sets out a three-stage arc — **a model, a self-learning model, and a hive mind** — and is explicit about which stages exist today. The first two are running. The third is a design direction with no implementation, and this paper says so rather than letting a roadmap read as a product.

On graphene, specifically

Stiffness per kilogram is what a robot limb actually cares about, and on that measure every conventional structural metal is the same metal: aluminium, titanium and steel all sit within 6% of **25 MN·m/kg**. Graphene-enhanced CFRP sits at **45.2** — **77% above the entire metals plateau**, which at equal stiffness is an axial member 44% lighter. On the thermal side a graphene heat-spreader film runs **7.8×** aluminium in plane. Those are the two numbers that justify a graphene programme in a humanoid, and section 7 derives both from the same material tables the CAD is sized against.

Section 7 also states, in the same place, that **none of those materials has been measured on a coupon in our own programme**. The system is specified and applied to 89 parts today; it is not validated, and we would rather print that than let a reader assume otherwise.

Who this is for

Engineering leaders evaluating humanoid platforms, materials groups considering graphene in load paths, and policy readers who want to know what a sovereign robotics capability actually requires. It assumes no familiarity with our stack. Readers who came for the materials work can go straight to section 7.

1 The real bottleneck

Every serious humanoid programme eventually discovers the same thing: the hard part is not building a robot that can move, it is building one that knows what to do.

The mechanical problem is largely solved and increasingly commoditised. Actuators with respectable torque density are purchasable. Open humanoid platforms — Asimov, Berkeley Humanoid Lite, Fourier's N1 — publish CAD, URDFs and control stacks. A competent team can have a walking chassis in months.

What that team then discovers is that the chassis is perhaps fifteen per cent of the problem. The remaining eighty-five is **experience**: the millions of interactions from which a policy learns that a crate is heavier than it looks, that a person walking towards you will keep walking, that a door handle turns before it pulls. Collecting that on real hardware is slow, expensive, and breaks robots.

This is why the countries and companies ahead in humanoids are ahead on simulation and data, not on mechanism. And it is why our answer starts with the simulator rather than the robot.

1.1 What sovereign means here

A sovereign capability is one you can still operate when a supplier, a licence or a jurisdiction changes its mind. In practice that means: weights you hold, a simulator you can modify, a bill of materials with more than one possible supplier per line, and no runtime dependency on an API you do not control. Every architectural choice below is downstream of that.

2 Stage one — a model

A world model with a body. Frozen foundations, trained heads, and one ordered contract between the two.

2.1 The architecture

MOTHER EXO is built as a **frozen core with trained heads**. A frozen language-reasoning core and a frozen vision transformer supply general competence; task-specific heads are trained on top. Nothing retrains the foundation.

That choice is not about compute economy, though it helps. It is about **auditability**. When a head misbehaves you can retrain that head, compare it against its predecessor, and roll back — because the thing underneath it did not move. A monolithic model that is fine-tuned end to end gives you a single artefact whose regressions are impossible to localise.

The heads span perception, understanding, spatial relationships, world modelling, tracking, memory, reasoning, social awareness, expression and motor control. Each has its own weight version and its own evaluation.

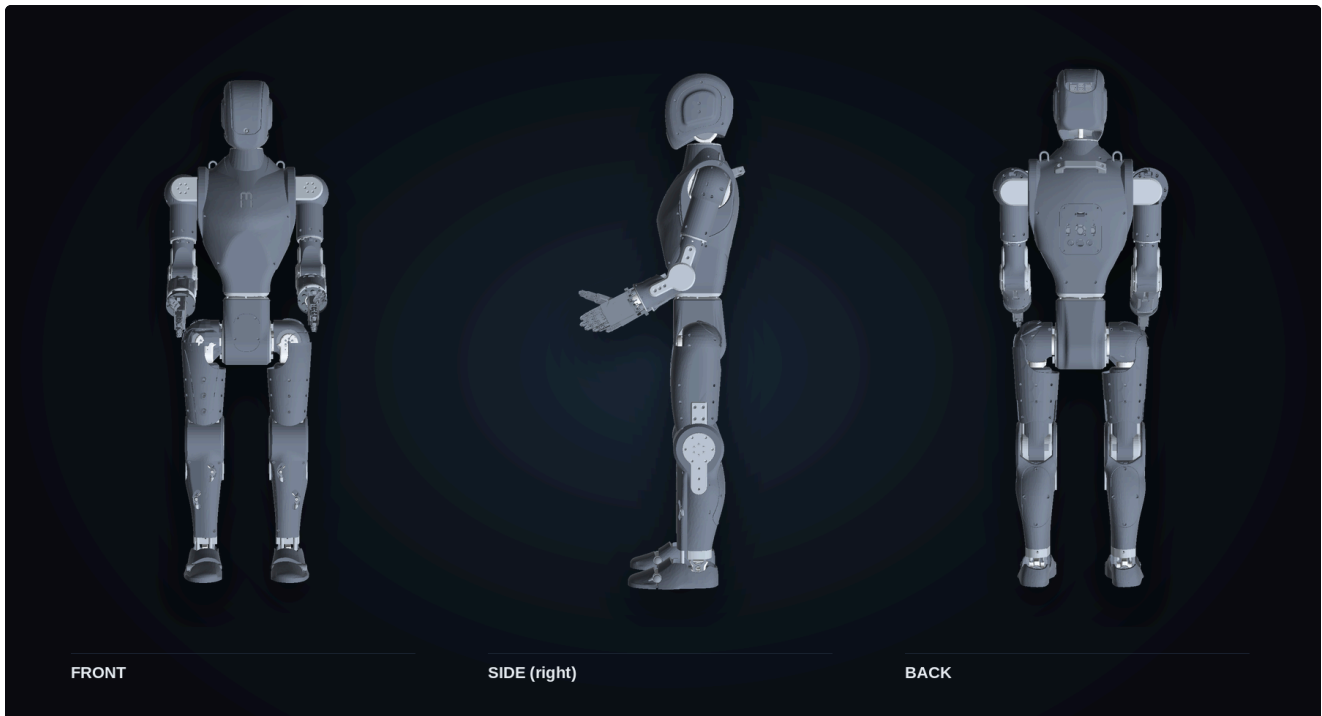
2.2 The contract between brain and body

A world model that cannot act is a classifier. The link is a single ordered list of 27 actuated degrees of freedom which simultaneously defines the action vector, the proprioception block order, and — as we found — the actuator purchase quantity.

WHY ONE LIST MATTERS

Our purchase BOM originally counted actuators from the mechanical CAD's actuation schedule instead of from this contract. The schedule is shaped by the baseline URDF, which models the neck as two *fixed* joints and predates our three-axis waist. It listed 23 drives for a 27-DoF robot — and totalled cleanly, because 23 real actuators at real prices is exactly what a correct BOM looks like. Deriving the count from the contract instead makes the error impossible: the contract is what defines how many there are.

2.3 The body



MOTHER EXO general arrangement. The body is the Asimov v1 open humanoid; the waist, hands and material system are MSAI's.

We took an open-source baseline deliberately. There is no advantage in re-drawing a femur, and considerable advantage in inheriting a platform others are also testing. Our engineering went where the baseline ran out.

3 Where the baseline ran out

Two modules, both driven by manipulation rather than locomotion.

3.1 A hand that opposes



MEX-H1. Fifteen joints on six drives, tendon-driven, with the drives in the forearm.

Open humanoid platforms overwhelmingly ship grippers. A gripper is the right answer for pick-and-place and the wrong answer for a world model, because most of what a hand teaches you about the world comes from **contact variety** — and a parallel jaw has one contact mode.

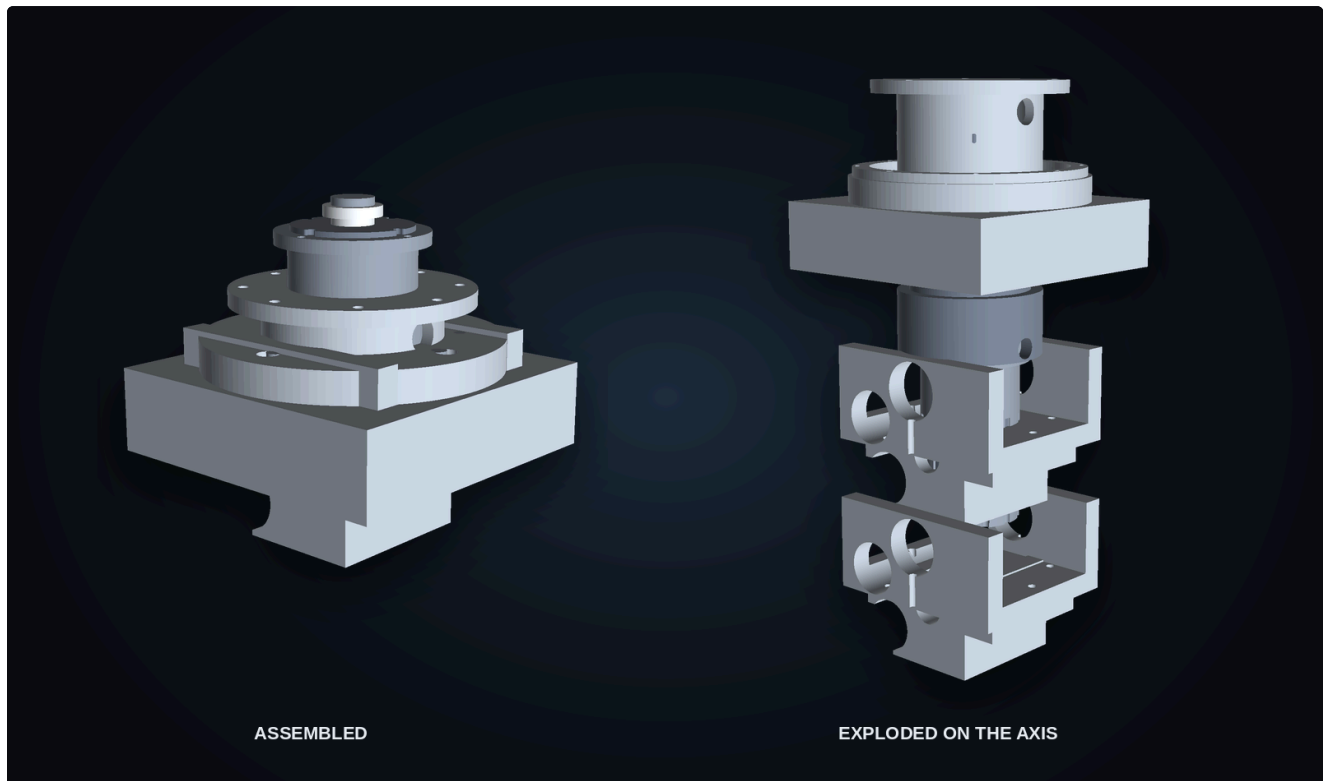
MEX-H1 has 15 joints driven by 6 actuators through tendons, with the drives in the forearm so the hand carries no motor mass. The interesting engineering is the thumb. A human carpometacarpal joint is a saddle: two axes whose combination rotates the pad to face the fingers. Our first design had one hinge, and no amount of tuning produced a pinch — the thumb's flexion plane stayed parallel to the fingers' and the two chains curled past each other forever.

Sweeping the pose space with forward kinematics showed a pinch pose **did** exist inside the segment lengths; the joint simply could not reach it. Rather than add a seventh drive, we solved for a single mount frame and hinge axis that reaches a square-on pinch. Verified in MuJoCo on both hands: pad gap 1.74 mm with the pad normals opposed at -0.966.

WHAT WE HAVE NOT SOLVED

The hand **reaches** a pinch. It cannot yet **close** one on an object: with both tendons at full tension the fingers run past each other at 4.2 mm. Closing on a real object needs tendon tensions modulated against the tactile pads — a controller nobody has written yet. We report this because a hand that poses for a photograph and cannot grip is a well-understood failure mode in this industry, and the distinction matters.

3.2 A waist that turns forever



MEX-W1 and MEX-W2, assembled and exploded. A crossed-roller bearing carries the upper body; a hybrid slip ring carries power and network up the hollow axis.

Humanoid torsos almost universally have a limited yaw joint, because a continuously rotating one has to solve cable wind. The consequence is that reaching behind requires stepping — which turns a manipulation problem into a locomotion problem, and makes training data far more expensive.

MEX-W1 makes yaw **continuous**. A crossed-roller bearing carries the upper body so the motor supplies torque only; a hybrid slip ring in the hollow shaft carries 20 circuits — 48 V, Ethernet, CAN-FD and a hard-wired E-stop loop — so nothing winds. Torso-local compute keeps that circuit count low: the waist carries a network, not every sensor pair.

The slip ring occupying the axis is what shapes the drive train. The reduction has to work around the bore, which forces a ring-and-pinion stage on top of the strain wave: 183.3:1 total, and a real top speed of 3.43 rad/s rather than the 12.57 the URDF had inherited from an actuator-family default. That correction matters more than it sounds: left uncorrected, every policy trained in simulation would have learned to command a speed the hardware cannot reach.

Beneath it, MEX-W2 adds pitch ($\pm 30^\circ$) and roll ($\pm 25^\circ$). These are not the yaw module on its side. The yaw axis is vertical so gravity does not act on it; pitch and roll are horizontal, and the sizing driver becomes **holding** torque — 24.0 N·m at full lean — rather than peak. Hence a 100:1 reduction that will not back-drive, and a power-off brake on each axis.

4 Stage two — self-learning

Continuous learning is easy. Continuous learning that does not destroy what it already knew is the engineering.

A model that trains on new material every night will, without countermeasures, quietly lose old skills. Catastrophic forgetting is not a corner case; it is the default behaviour of gradient descent on a shifting data distribution. Three pieces work together to prevent it:

The replay archive

Every instruction the system has ever been taught is archived and indexed by head. Each night's training replays a sample of that archive alongside the day's new material. Experience replay is the mechanism that stops new learning from overwriting old.

The retention guard

After teaching anything new, a sample of **previously learnt** instructions is re-evaluated. If any head's pass rate falls below its retention floor, the guard flags forgetting and the promotion gate fails until a replay-heavy retrain restores it. A model that has regressed cannot be promoted merely because the new material scored well.

The nightly trainer

Gathers the day's instructions plus a replayed archive sample, trains each affected head independently, bumps that head's weight version, and queues the job. Training runs only with **instructor approval** — continuous learning stays under human guidance by design, not by convention.

4.1 Cheap data and scarce data, separated

Heads divide by what they can learn from. Perception, spatial reasoning and reasoning have structure learnable from a large **unlabelled** archive by predicting their own latents, so they are pretrained self-supervised. Locomotion, manipulation, safety, social and voice still need scarce labelled teleoperation and rubric data, and are trained supervised. Treating those two groups identically wastes the cheap data and starves the expensive heads.

A WEIGHT IS NEVER DELETED

A standing rule in this programme: a trained head is never wiped, overwritten or restarted. A head that has collapsed is fixed by training forward from its checkpoint, or by forking to a new head file with the original untouched. There is no 'fresh start' on an existing head. Continuity of trained weights is treated as an asset on the balance sheet, because that is what it is.

5 Stage three — the hive mind

Where this is going, stated as a direction rather than a feature.

NOT BUILT

There is no hive-mind implementation in our stack today. No multi-robot learning, no federated aggregation, no cross-unit weight sharing. This section describes a design direction and the problems it has to solve. We separate it from sections 2 and 4 deliberately, because a roadmap presented as a capability is the most common way robotics programmes lose credibility.

The argument for it is simple arithmetic. One robot accumulates experience at one robot's rate. A fleet of a hundred, each meeting different rooms, objects and people, accumulates a hundred times as much — provided that experience can be pooled without pooling everything else.

5.1 What has to be solved first

- **Aggregation without raw data.** A robot in a customer's building cannot ship its camera feed anywhere. The unit of sharing has to be a learned update, not an observation.
- **Retention across a fleet.** The retention guard described above governs one model's history. A fleet needs the same guarantee across units that have seen genuinely different worlds — and a unit that has specialised usefully must not be flattened by the average.
- **Provenance.** If a behaviour regresses after a fleet update, you must be able to say which unit's experience introduced it. That is a data-lineage problem before it is a machine-learning problem.
- **Divergence as a feature.** A hospital robot and a warehouse robot should not converge. The interesting architecture keeps a shared trunk and lets deployment-specific heads differ.

Our frozen-core, trained-head architecture is chosen partly with this in mind: it gives a natural unit of exchange. Sharing a head is tractable in a way that sharing a monolithic fine-tune is not. That is an argument for the architecture, not evidence that the hive mind works.

6 The Digital SIM

The platform that lets a small programme accumulate experience at the rate of a large one.



The physics twin — geometry read out of the compiled MuJoCo model and posed by its own forward kinematics. One robot, not two.

The MOTHER EXO Digital SIM is a real-physics twin of the machine: 26 rigid bodies, 24 joints, 48 collision geoms and 959,036 mesh triangles, running on MuJoCo. It is not a visualisation. Joint limits, torque envelopes, link inertias, contact and self-collision are the ones the hardware has, because they are read from the same URDF the hardware is built to.

6.1 One robot, not two

The failure mode of digital twins is drift: the simulator's robot and the workshop's robot become different machines, and everything learned in the first stops transferring to the second. We attack that structurally. There is a single source for geometry, a single contract for degrees of freedom, and figures in this paper are rasterised from the same data the simulator compiles — including the illustrations above. A render that disagreed with the model would be a render this toolchain cannot produce.

6.2 Training on what the model will actually see

An early and instructive mistake: our vision heads are trained on real imagery, but the simulated world was built from primitive shapes. A policy learning perception in that world learns to recognise a texture-less grey box — which no camera will ever show it. The simulated environments now carry reference imagery attached to objects, so the perception heads see something closer to what they were trained on.

6.3 Why this is the catch-up mechanism

Simulation is the only part of this stack where a small team can be competitive with a large one on equal terms. Compute is purchasable, physics engines are open, and the marginal cost of the ten-thousandth simulated hour is

near zero. It is where a national or independent programme can close a gap that would be unclosable in hardware alone.

7 Graphene in robotics

The material argument, in numbers — and the exact point at which our evidence runs out.

7.1 Every structural metal is the same metal

A limb does not care about stiffness. It cares about **stiffness per kilogram**, because every gram in a moving link is a gram the actuators must accelerate and the structure must carry. On that measure the conventional options are indistinguishable:

MATERIAL	MODULUS (GPA)	DENSITY (KG/M ³)	STIFFNESS PER KG (MN·M/KG)
Aluminium 6061-T6	69	2,700	25.6
Aluminium 7075-T6	71.7	2,810	25.5
Titanium Ti-6Al-4V	113	4,430	25.5
Stainless 316L	193	8,000	24.1

Computed from `mother_exo_sim/cad/materials.py` — the same table the CAD is sized against.

Titanium is 64% stiffer than aluminium and 64% denser. Steel is nearly three times stiffer and three times denser. Swapping between them buys strength, temperature capability or cost — it does not buy stiffness per kilo. Every structural metal sits on the same plateau at roughly **25 MN·m/kg**, and a humanoid built from any of them is fighting the same mass budget.

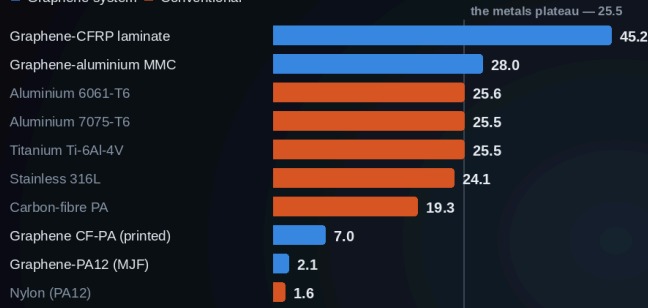
Graphene-enhanced CFRP sits at **45.2**.

Every structural metal is the same metal

Stiffness per kilo is what a limb actually cares about — and aluminium, titanium and steel all land within 6% of each other.

SPECIFIC STIFFNESS E/ρ · MN·m/kg

■ Graphene system ■ Conventional

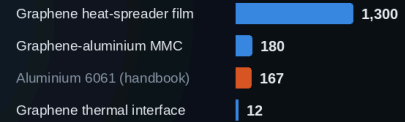


GRAPHENE-CFRP vs ALU 7075

+77%

stiffness per kilogram, at equal mass
An axial member carrying the same load is 44% lighter.

IN-PLANE THERMAL CONDUCTIVITY · W/m·K



7.8× aluminium on the drive and compute heat path.
Aluminium is a handbook figure, carried for comparison only.

WHERE IT ALREADY IS, TODAY

Not a proposal — these are parts in the shipped CAD, each carrying its own material key.

89

parts on a graphene material

51

surfaces with a graphene finish

16

materials across 7 systems

12,412

e-skin taxels in the sensor suite

DESIGN RULE 1

Monolayer graphene — 1,000 GPa, 4,000 W/m·K — appears nowhere in the chart above. No part is ever sized from it. Structural models use the M1 laminate and MMC rows only.

Source: mother_exo_sim/cad/materials.py and cad/graphene_materials.py — the same tables the CAD is sized against.

WHAT WE HAVE NOT DONE

Of 16 materials in the system: 4 rest on vendor data, 11 on published literature, 1 is a theoretical reference. NONE has been measured on a coupon in this programme. Phase 2 has not started.

Specified and applied everywhere. Not validated.

The case in numbers. Specific stiffness and in-plane thermal conductivity, computed from the repository's own material tables.

Source: cad/materials.py and cad/graphene_materials.py

WHAT +77% ACTUALLY BUYS

At equal stiffness, an axial member in graphene-CFRP masses **44% less** than the same member in 7075. On a machine where the waist already carries 18.36 kg of upper body at 267 mm, mass removed from the limbs compounds: smaller holding torques, smaller brakes, smaller actuators, less battery to move them.

Bending-dominated sections do better still, because mass there scales with $\rho/E^{(1/3)}$ rather than ρ/E .

7.2 And it is not only structure

The second win is thermal, and it is larger in ratio than the structural one. A humanoid's real thermal problem is getting heat out of drive baseplates and compute into something that will radiate it, without adding mass to a moving link. A graphene heat-spreader film runs at **1,300 W/m·K** in plane against aluminium's ~167 — **7.8×**, at a fraction of the thickness.

Four honest opportunities, then, and we specify graphene at all four:

- **Structure (M1).** Graphene-CFRP and graphene-aluminium MMC, the only materials in our library that clear the metals plateau.
- **Thermal (M3).** Spreaders and interface material on the drive and compute heat paths.
- **Sensing (M4).** Piezoresistive composites make a skin that is thin, conformable and cheap per square metre. The design sensor suite calls for **12,412 taxels**; no conventional sensor reaches that on cost.
- **Tribology (M6).** A continuously rotating waist turns for as long as the robot is powered, so the sliding interfaces are worth more here than anywhere else on the machine.

7.3 Where graphene does not go

Marketing claims for graphene rarely survive contact with a load path, so the system carries design rules that constrain it. The first exists purely to stop the headline numbers being used:

DESIGN RULE 1

Monolayer graphene figures are reference only — no part is ever sized from them. Structural models use the M1 laminate and metal-matrix composite rows.

Monolayer graphene is 1,000 GPa and 4,000 W/m·K. Those numbers describe one perfect sheet, and they appear nowhere in the chart above.

The others follow from the same instinct: graphene is specified only where it beats the incumbent on **system** terms — mass, watts, cost or volume — and where steel, titanium, aluminium or copper win, they stay. Nothing relies on unsupported graphene in compression, or on through-plane conduction, because graphene carries in-plane loads and in-plane heat and is unremarkable in the other direction.

7.4 It is already in the machine

This is not a materials proposal attached to a metal robot. Every part in the shipped CAD carries a material key and a surface finish, decided by one question — *does it slide?*

89

PARTS ON A GRAPHENE MATERIAL

51

SURFACES WITH A GRAPHENE FINISH

16

MATERIALS, 7 SYSTEMS

12,412

E-SKIN TAXELS SPECIFIED

WHERE	MATERIAL	PARTS
Hand and waist structure, printed	Graphene-PA12 (MJF)	56
Highly loaded fittings	Ti-6Al-4V, graphene-coated	23
Structural plates, prototype route	Graphene + chopped-CF PA	10
Anything that slides or rolls	Graphene tribological coating	29
Environmental protection	Graphene barrier coating	12
Fingertips	Graphene e-skin, piezoresistive	10

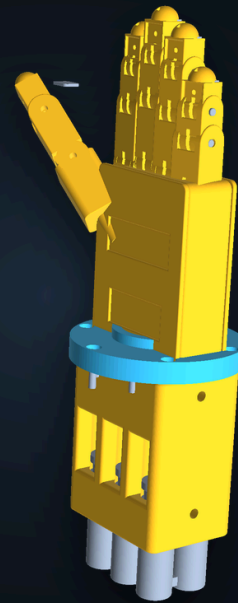
Counted from the shipped placement records and the waist graphene map — not from a plan.

That vocabulary is shared by implementation rather than by convention, so the hand and the waist cannot drift onto different material systems. And the surface language itself was measured, not chosen: across all 168 baseline fabrication files, MJF parts carry R1.5 edge breaks, sintered steel R1.0, machined aluminium essentially none. Applying the same rule to our own modules is why they read as one machine rather than as bolt-ons.

- Graphene + chopped-CF PA (printed)
- Graphene-PA12 (MJF)
- Ti-6Al-4V with graphene coating

Surface finishes

- Graphene barrier coating
- Graphene e-skin (piezoresistive)
- Graphene tribological coating



Every hand part coloured by its own recorded material key.

7.5 What we cannot claim

Everything above is a specification. None of it is a measurement we have taken.

THE LINE OUR EVIDENCE STOPS AT

Of sixteen materials, **four** rest on vendor data, **eleven** on published literature, and **one** is a theoretical reference. **None has been measured on a coupon in this programme.** Phase 2 of our seven-phase materials plan — the phase that replaces every literature value with a measured one — has not started.

So the material system is specified, applied consistently to every part, and integrated with the CAD. It is **not validated**. Anyone claiming validated graphene structural performance in a humanoid today, ourselves included, is describing literature.

We publish that because the alternative is worse. A material claim that cannot say where its evidence stops is not a claim an engineer can build against, and a supplier who will not draw that line is a supplier you cannot plan a structure around. The +77% above is a literature figure with a named source and a test plan attached. That is a different object from a marketing number, and the difference is the whole point of this section.

7.6 The programme that closes it

PHASE	GOAL	STATUS
1. Materials database	Measured, not marketing, properties for every row above	in progress
2. Coupon testing	Replace every literature-confidence number with measured data	not started
3. Structural components	Bone → joint → limb → torso, built and tested to failure	not started
4. Smart skin	Tactile and strain network at full body coverage	not started
5. Thermal system	GB10 and drive thermal paths validated under duty	not started
6. Energy	Battery + supercapacitor hybrid under humanoid duty profile	not started
7. MOTHER EXO v1 integration	Full material stack in the humanoid	not started

Phase 3's exit criterion is a 500–800 mm limb test article, built and tested to failure — the demonstrator that de-risks the material stack before it touches the humanoid.

Phase 2 is the one that matters and the one that is next. Until tensile, ILSS, fatigue, conductivity and environmental results exist for M1 through M4, every structural figure in this paper carries the word *literature* beside it — and we would rather print that word than quietly drop it.

8 Open source and in-house

What we took, what we built, and why the line falls where it does.

LAYER	SOURCE	RATIONALE
Body structure	Asimov v1 (open)	No advantage in re-drawing a femur; real advantage in a platform others test too
Physics engine	MuJoCo (open)	Mature contact dynamics; modifiable; no runtime licence
Locomotion reference	Berkeley Humanoid Lite, Fourier N1 (open)	Published control stacks worth learning from
Hand	MSAI	Open platforms ship grippers; a world model needs contact variety
Waist	MSAI	No open humanoid solves continuous torso rotation
Material system	MSAI	Graphene integration with the CAD does not exist off the shelf
Digital SIM platform	MSAI	The training environment is the competitive asset
Model architecture and heads	MSAI	Frozen core, trained heads, held weights — sovereignty requires it

The pattern is consistent: take the commoditised, build the differentiating. Mechanism is commoditised. Manipulation hardware, the training environment and the model are not.

9 Economics and availability

What a unit costs to build, on a bill of materials derived from the CAD.

£36,287

PARTS PER UNIT

£42,530

BUILD COST

£52,362

TOTAL COST PER UNIT

2027

AVAILABILITY

Those figures come from a bill of materials generated from the mechanical CAD — actuator counts from the degrees-of-freedom contract, part counts from the fabrication manifest, module masses from measurement snapshots. They are not a target cost worked backwards from a price.

HONEST ABOUT THE PRICE BASIS

Not one line of that bill of materials is a held supplier quote. About a quarter rests on checked list prices; the balance is market-informed estimate or explicit placeholder. The cost of materials is therefore indicative, and we say so in our own internal budget in the same words.

MOTHER EXO is targeted for availability in **2027**. Between now and then the work that matters is not mechanical: it is the coupon programme that turns literature material values into measured ones, the grip controller that turns a hand that reaches into a hand that holds, and the accumulated simulated experience that turns a world model into a useful one.

9.1 Safety posture

Observe-and-advise, human-in-the-loop, **strike = 0** across every domain. Target selection, fire control and kill chains are refused as a matter of architecture, not policy. Continuous learning runs under instructor approval. Every destructive or autonomous-action path requires explicit human confirmation. These are constraints we build to, and we state them here because a humanoid programme that will not say where its limits are has not thought about them.

10 Conclusion

The humanoid race will not be won by whoever builds the best arm. It will be won by whoever accumulates useful experience fastest and keeps it. That is a statement about simulation, about continual learning that does not forget, and eventually about pooling experience across many machines.

MOTHER EXO is our position on that argument in hardware and software: an open baseline where mechanism is commoditised, our own engineering where it is not, a material programme honest about its own maturity, and a physics twin where the learning actually happens. Two of the three stages are running. The third is named as a direction, and will be reported as built when it is built.

CONTACT

MSAI Robotics · Media Stream AI · mediastreamai.com

MOTHER EXO — available 2027.